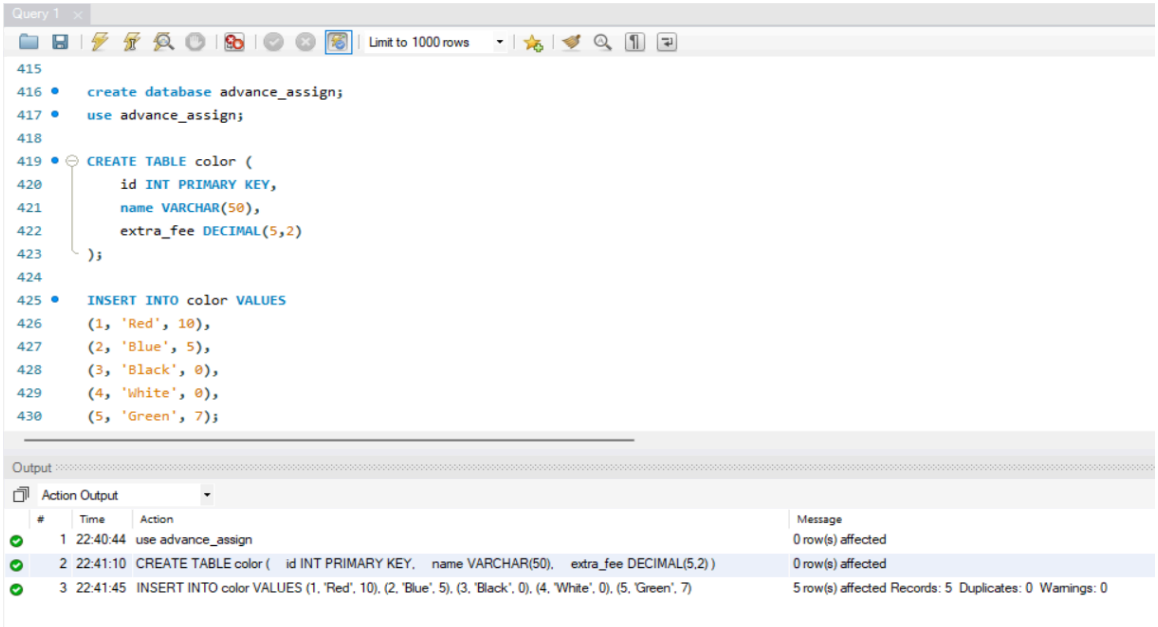
Kshitiz Gupta

ADVANCE SQL ASSIGNMENT

Introduction

In this assignment, I created sample tables and inserted data to test SQL queries. The queries were executed based on the given conditions and results were verified.

Creating table color and inserting values:-
create database advance_assign;
use advance_assign;



Creating table customer and inserting values:-

```sql
CREATE TABLE customer (
    id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    favorite_color_id INT,
    city VARCHAR(50),
    FOREIGN KEY (favorite_color_id) REFERENCES color(id)
);

-- this is sample data and values are not real
INSERT INTO customer VALUES
(1, 'Rahul', 'Sharma', 2, 'Mumbai'),
(2, 'Priya', 'Verma', 1, 'Delhi'),
(3, 'Amit', 'Patel', 3, 'Mumbai'),
(4, 'Sneha', 'Joshi', 4, 'Pune'),
(5, 'Karan', 'Mehta', 2, 'Mumbai'),
(6, 'Neha', 'Kapoor', 5, 'Delhi');
```

Output

Action Output

#	Time	Action	Message
1	22:44:26	CREATE TABLE customer (id INT PRIMARY KEY, first_name VARCHAR(50), last_name VARCHAR(50), ...	0 row(s) affected
2	22:44:32	INSERT INTO customer VALUES (1, 'Rahul', 'Sharma', 2, 'Mumbai'), (2, 'Priya', 'Verma', 1, 'Delhi'), (3, 'Amit', 'Patel',...	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0

Creating table category and inserting values:-

```sql
CREATE TABLE category (
    id INT PRIMARY KEY,
    name VARCHAR(50),
    parent_id INT,
    FOREIGN KEY (parent_id) REFERENCES category(id)
);

INSERT INTO category VALUES
(1, 'mens', NULL),
(2, 'mens_jeans', 1),
(3, 'mens_tshirt', 1),
(4, 'womens', NULL),
(5, 'womens_dress', 4),
(6, 'womens_top', 4);
```

Output

Action Output

#	Time	Action	Message
1	22:46:36	CREATE TABLE category (id INT PRIMARY KEY, name VARCHAR(50), parent_id INT, FOREIGN KE...	0 row(s) affected
2	22:46:39	INSERT INTO category VALUES (1, 'mens', NULL), (2, 'mens_jeans', 1), (3, 'mens_tshirt', 1), (4, 'womens', NULL),...	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0

Creating table items and inserting values:-
Creating table stores and inserting values:-

```sql
INSERT INTO item VALUES
(1, 'Blue Denim Jeans', 'M', 1500, 300, 2, 2),
(2, 'Black Slim Jeans', 'L', 1800, 400, 3, 2),
(3, 'Red Casual T-Shirt', 'S', 800, 200, 1, 3),
(4, 'White Summer Dress', 'M', 2000, 500, 4, 5),
(5, 'Green Cotton T-Shirt', 'XL', 900, 250, 5, 3);

CREATE TABLE store (
    id INT PRIMARY KEY,
    city VARCHAR(50)
);

INSERT INTO store VALUES
(1, 'Mumbai'),
(2, 'Delhi'),
(3, 'Pune'),
(4, 'Bangalore');
```

Output

Action Output

#	Time	Action	Message
1	22:47:44	CREATE TABLE item (id INT PRIMARY KEY, name VARCHAR(100), size VARCHAR(10), price DECI...	0 row(s) affected
2	22:47:49	CREATE TABLE store (id INT PRIMARY KEY, city VARCHAR(50))	0 row(s) affected
3	22:48:54	INSERT INTO item VALUES (1, 'Blue Denim Jeans', 'M', 1500, 300, 2, 2), (2, 'Black Slim Jeans', 'L', 1800, 400, 3, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
4	22:49:00	INSERT INTO store VALUES (1, 'Mumbai'), (2, 'Delhi'), (3, 'Pune'), (4, 'Bangalore')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0

Creating table orders and inserting values:-

```sql
CREATE TABLE `order` (
    id INT PRIMARY KEY,
    customer_id INT,
    item_id INT,
    item_cnt INT,
    store_id INT,
    order_channel VARCHAR(20),
    day_dt DATE,
    FOREIGN KEY (customer_id) REFERENCES customer(id),
    FOREIGN KEY (item_id) REFERENCES item(id),
    FOREIGN KEY (store_id) REFERENCES store(id)
);

INSERT INTO `order` VALUES
(1, 1, 1, 2, 1, 'Store', '2024-01-10'),
(2, 2, 3, 1, 2, 'Online', '2024-02-15'),
(3, 3, 2, 1, 1, 'Store', '2024-03-05'),
(4, 1, 3, 1, 1, 'Online', '2024-03-20'),
(5, 5, 1, 1, 1, 'Store', '2024-04-01');
```

Output

Action Output

#	Time	Action	Message
1	22:55:07	CREATE TABLE `order` (id INT PRIMARY KEY, customer_id INT, item_id INT, item_cnt INT, store_i...	0 row(s) affected
2	22:55:21	INSERT INTO `order` VALUES (1, 1, 1, 2, 1, 'Store', '2024-01-10'), (2, 2, 3, 1, 2, 'Online', '2024-02-15'), (3, 3, 2, 1, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

Query 1:

```
SELECT
   i.name AS item_name,
   col.name AS color_name,
   cust.first_name,
   cust.last_name
FROM `order` o
JOIN customer cust ON o.customer_id = cust.id
JOIN item i ON o.item_id = i.id
JOIN color col ON i.color_id = col.id
JOIN category cat ON i.category_id = cat.id
JOIN store st ON o.store_id = st.id
WHERE
   cust.city = 'Mumbai'
   AND o.order_channel = 'Store'
   AND cat.name = 'mens_jeans'
   AND cust.favorite_color_id = i.color_id
ORDER BY col.name ASC;
```

```sql
SELECT
    i.name AS item_name,
    col.name AS color_name,
    cust.first_name,
    cust.last_name
FROM `order` o
JOIN customer cust ON o.customer_id = cust.id
JOIN item i ON o.item_id = i.id
JOIN color col ON i.color_id = col.id
JOIN category cat ON i.category_id = cat.id
JOIN store st ON o.store_id = st.id
WHERE
    cust.city = 'Mumbai'
    AND o.order_channel = 'Store'
    AND cat.name = 'mens_jeans'
    AND cust.favorite_color_id = i.color_id
ORDER BY col.name ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

item_name	color_name	first_name	last_name
Black Slim Jeans	Black	Amit	Patel
Blue Denim Jeans	Blue	Rahul	Sharma
Blue Denim Jeans	Blue	Karan	Mehta

Result 12

Output

Action Output

#	Time	Action	Message
1	23:01:24	SELECT i.name AS item_name, col.name AS color_name, cust.first_name, cust.last_name FROM `ord...	3 row(s) returned

Explanation:

I joined multiple tables to get item, customer, and color details. I filtered the data for Mumbai customers who made store purchases in the mens_jeans category. I also matched the item color with the customer's favorite color and sorted the result by color.

Query 2:

```
SELECT
    cust.first_name,
    cust.last_name,
    col.name AS favorite_color
FROM customer cust
LEFT JOIN `order` o ON cust.id = o.customer_id
LEFT JOIN color col ON cust.favorite_color_id = col.id
WHERE o.id IS NULL;
```

```
SELECT
    cust.first_name,
    cust.last_name,
    col.name AS favorite_color
FROM customer cust
LEFT JOIN `order` o ON cust.id = o.customer_id
LEFT JOIN color col ON cust.favorite_color_id = col.id
WHERE o.id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	favorite_color
Sneha	Joshi	White
Neha	Kapoor	Green

Result 15

Output

Action Output

#	Time	Action	Message
1	23:04:38	SELECT cust.first_name, cust.last_name, col.name AS favorite_color FROM customer cust LEFT JOIN `...	2 row(s) returned

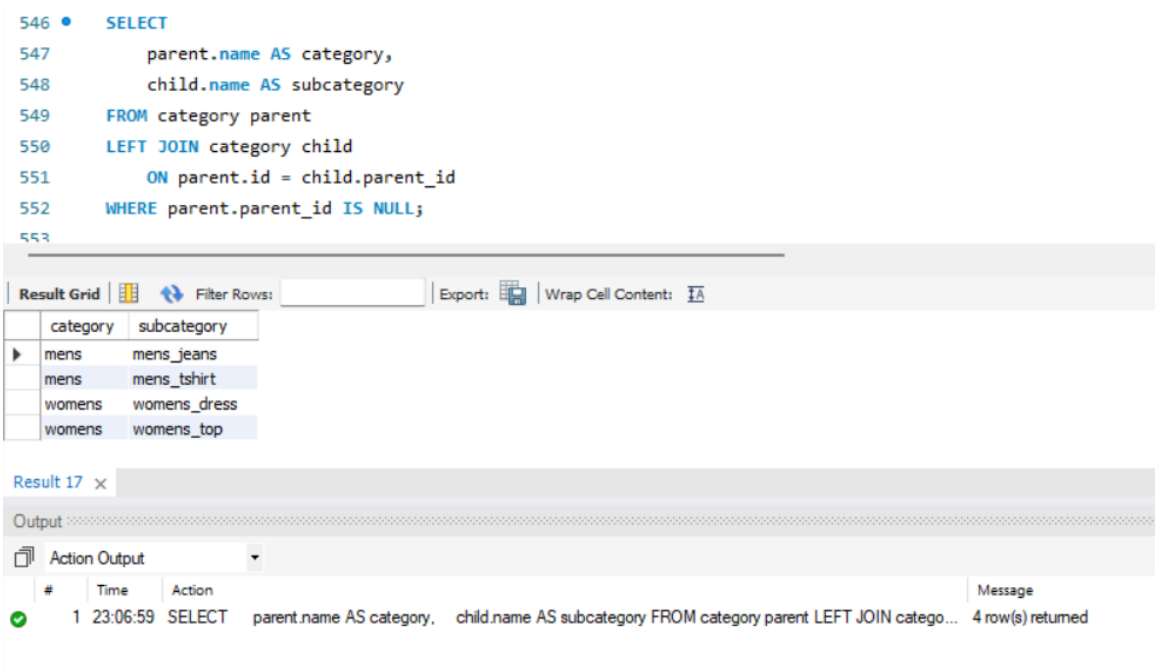
Explanation:

I used LEFT JOIN to include all customers. Then I filtered those whose order ID is NULL, which means they have not made any purchase. I also joined the color table to display their favorite color.

Query 3:

```
SELECT
    parent.name AS category,
    child.name AS subcategory
FROM category parent
LEFT JOIN category child
    ON parent.id = child.parent_id
WHERE parent.parent_id IS NULL;
```

Output Screenshot:



Explanation:

In this query, I joined the category table with itself to get main categories and their subcategories. Main categories are identified by NULL parent_id. LEFT JOIN ensures that even if a category has no subcategory, it is still displayed.